BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

LESSON 153 · Paper 1: Data Representation

Name: _____ Class: _____ Date: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

Do Now · Recall previous learning

Q1. What does the MDR hold?

Q2. grid = [[11,12],[21,22]]. With indexes starting at 0, what is grid[1,0]?
A. 21
B. 11
C. 12
D. 22

Q3. “Changing a variable name automatically fixes every logic error.” Correct this. Explain why.

Q4. Two functions each have a local total. Several procedures share a global score. Correct both claims. Explain why. (a) All subprograms can read every local variable. (b) Only one subprogram can read a global variable.

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I Do · Read new knowledge

Key vocabulary

Binary place values — Use 128,64,32,16,8,4,2,1 for unsigned eight-bit numbers.

Hexadecimal — Group binary digits into nibbles of four bits.

Image size — Width×height×colour depth yields uncompressed bits.

Sound size — Samples/second×seconds×bits/sample yields mono uncompressed bits.

New learning

Binary place values: Use 128,64,32,16,8,4,2,1 for unsigned eight-bit numbers.

Hexadecimal: Group binary digits into nibbles of four bits.

Image size: Width×height×colour depth yields uncompressed bits.

Sound size: Samples/second×seconds×bits/sample yields mono uncompressed bits.

Worked example

Convert eight-bit 10110110 to denary and hexadecimal.

1. Add values: 128+32+16+4+2=182.

2. Split 1011 and 0110.

3. Convert to B6 hexadecimal.

4. Check 11×16+6=182.

We Do · Practise with guidance

Guided 1

Calculate the bits in a 60×40 bitmap with four-bit colour depth.

Worked method: Divide eight to get 1200 bytes if requested.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

Code First · Predict → Write → Test → Improve

Correct the raw data size conversion from bits to bytes.

```
width = 20
height = 10
depth = 8
bytes = width * height * depth
print(bytes)
```

iterate

Question 1

Convert hex 2A to denary. [3 marks]

Question 2

Convert eight-bit binary 11001100 to hex. [2 marks]

Question 3

Calculate mono sound bytes at 1000 samples/s, 4 s, 8 bits/sample. [4 marks]

Exit Ticket · Answer before checking

One hexadecimal digit can represent four binary bits.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

Do Now Recall previous learning | **I Do** Read new knowledge | **We Do** Practise with guidance | **You Do** Work independently | **Review & Improve** Check understanding

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

Page 1 onwards · Do Now

Q1. Data or an instruction being transferred to or from memory.

Q2. A — 21

Q3. Clear names show what values mean and help people read and change code.

Q4. (a) Each local total is used only in its own scope. (b) A global score can be used across the program. This risks unwanted changes. Use arguments and return values to make changes clear.

I Do / We Do pages · Guided practice

Q1. Divide eight to get 1200 bytes if requested.

You Do pages · Code First

```
width = 20
height = 10
depth = 8
bytes = width * height * depth DIV 8
print(bytes)
```

Expected result: 200

You Do pages · Independent questions

Q1. 2×16+10=42.

Q2. CC.

Q3. 1000×4×8÷8=4000 bytes.

Exit Ticket · Model answer

True

Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

1. Copy the whole original question:

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

2. Write a complete new answer in GREEN PEN. Show working if needed: